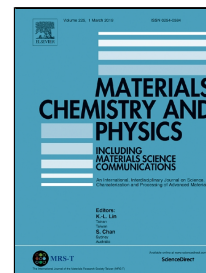


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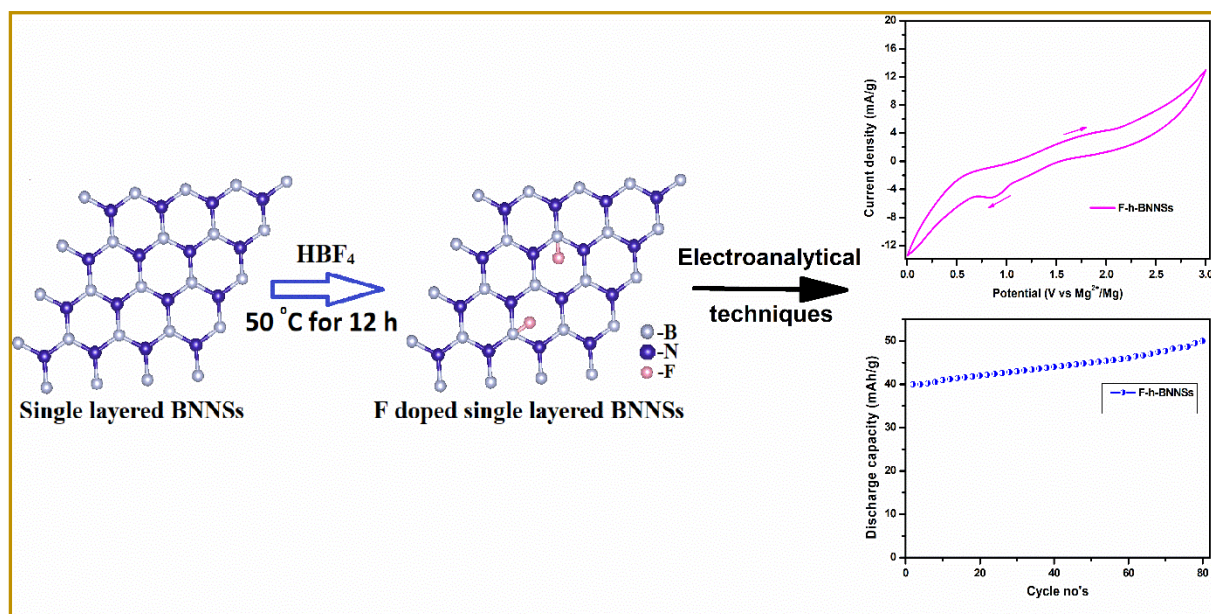
An Effective Performance of F-Doped Hexagonal Boron Nitride Nanosheets as Cathode Material in Magnesium Battery

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Graphical abstract



1. Introduction

The growth in demand for the portable battery with high energy density system is exerting pressure for the development of advanced new batteries. Magnesium (Mg) battery is the most suitable battery for the modern world since it has high safety standards compared to lithium and relatively low raw material costs [1-4]. Metallic Mg has negative electrode potential (-2.37 V) and high theoretical volumetric capacity (3833 mAh.cm⁻³) [5-7] however, it is highly unlikely to expect the reversible magnesium intercalation and de-intercalation into host material due to the strong polarizability of magnesium ions [8].

The study of the non-aqueous electrolyte has developed promptly since Aurbach et al. [9] prepared completely ethereal solvents of Grignard reagents RMgX (R = alkyl, aryl groups; X = halide groups), and Mg complexes reported in 2000. However, these procedures are very difficult in absorption-desorption of magnesium species, where the oxidation and reduction processes were occurred [10]. Therefore, it is required to find an appropriate electrolyte for the prospect of magnesium battery. So, imidazolium-based ionic liquid showed a broad range of characteristics such as a great thermal stability, lower vapor pressure, non-flammability, a good conductivity, a wide electrochemical window and good electrochemical stability [11]. Therefore, ionic liquid 1,3 dibutyl imidazolium bromide [DBIm]Br have a good performance possibility in the rechargeable battery.

Controlled doping in nanomaterials is very important phenomena since material character and application are exclusively depend on controlled doping, such materials are useful for energy storage system [12, 13]. Graphene exfoliated from graphite powder is one of the best example in which graphene is doped by boron (B) and nitrogen (N) elements this led to changes in properties and application [14-16]. The hexagonal boron nitride nanosheets (h-BNNSs) are analogues to graphene, [17-19] which is also called as white graphene, in which alternating N

and B atoms entered in place of C atoms, however, when it is compared to graphene it possess weak Van der Waals forces of attraction in between the BN layers and it has strong ionic bond as well as approximate band gap of h-BN around $\sim 4\text{-}6\text{ eV}$ [20, 21]. Generally, BN is insulating nature and does not show any electrochemical performance. Controlled doping method helps to minimize the band gap of BN which can improve electrical properties and its suitable material for many fields due to high thermal conductivity, chemical stability, and high mechanical properties [21, 22].

Our study particularly focussed on minimizing the band gap of h-BNNSs by doping some characteristic elements which lead to a considerable change in its conducting properties. Doping favourably takes place at the more vulnerable sites, so it will be much simpler to achieve doping experiment with a few layered h-BNNSs. Many previous studies have shown that the different methods are known to prepare mono or few layered h-BNNSs, [23, 24] but fluorine (F) doping is phenomenal which leads to a new pathway to control the electrochemical properties of h-BN. Since F is a more electronegative element and it has more valence electrons, when compared to N and B. Doping fluorine into selected nanomaterial should consistently hold a p-type semiconducting property at low coverages and a conducting behavior at high coverages [25, 26]. In previous reports shown that the probable functions of doping F into h-BN nanotubes and h-BN nanosheets [27, 28]. Tang et al. [25] reported that the electrical conductivity of BN nanotubes was studied but none of the previous studied does not exist in the practical application. With due considerations made above, we present our report on the exfoliated/fluorinated h-BN as cathode and magnesium as an anode for rechargeable Mg battery.

In this work, we prepared fluorinated hexagonal boron nitride nanosheets (F-h-BNNSs) by exfoliation of boron nitride through F doping into nanosheets of boron nitride via chemical solution process with fluoroboric acid. The fluorinated BN nanosheets (F-h-BNNSs) displays

superior electrochemical performance with a good degree of reversibility, good coulombic efficiency and good rate performance for magnesium battery.

2. Experimental Section

2.1 Chemicals

Born nitride powder, Isopropyl alcohol (IPA), Fluoroboric acid (48wt.% HBF_4 in H_2O), 1-butyl imidazole, n-butyl bromide, acetone, deionized water, polyvinyl difluoride (PVDF), super-p carbon black, N-methyl pyrrolidinone (NMP) and magnesium powder were purchased from Sigma Aldrich Pvt.Ltd.

2.2 Exfoliation of bulk BN

In exfoliation method, the bulk h-BN powder (0.30g) was dispersed in isopropyl alcohol (IPA) solution in a 50 ml RB flask and the resultant solution was heated up to 50 °C for 24 h with constant stirring. Thereafter, the solution was ultrasonicated for 24 h in a sonic bath. The obtained solution was left for 2 days and subjected to centrifugation at 5000 rpm for 30 min. Later, the sediment is collected by removing the supernatant solution. The obtained exfoliated product was washed with acetone solvent to remove remaining IPA solvent and dried at 60 °C for 12 h. Finally, a white powder containing layered h-BN nanosheets was obtained as shown diagrammatically in Fig. 1(a).

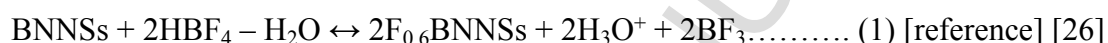
Figure 1

2.3 Fluorination of h-BNNSs

In fluorination process, as shown in Fig. 1(b), the exfoliated h-BN (0.30g) and fluoroboric acid (48wt.% HBF_4 in H_2O -20 mL) [21] were mixed in 50ml RB flask. Then the resultant mixture was heated up to 50 °C for 12h with constant stirring, the resultant product was cooled at

atmospheric temperature. Further, centrifugation followed by washing with de-ionized water and dried at 90 °C for 12h. Finally, the fluorinated product is known as fluorinated h-BNNSs (F-h-BNNSs).

The F doping mechanism as shown in single layered BN nanosheet in Fig. 1(b), in general, an electronegative species (F) donates an electron and form an ionic bond with the single-layered BN networks [26] (represented by 2 B-F bond formation, in equation (1)). If the fluorine (F) atoms binding on the surface of h-BNNSs, it will be unstable, and the electrical conductivity does not improve [25]. So, we decide that the F was doped into the layered BN nanosheets as represented in the following equation (1).



2.4 Synthesis of ionic liquid electrolyte

Scheme 1 represents the imidazole related ionic liquid 1,3-dibutylimidazolium bromide [DBIm]Br. 1,3-dibutylimidazolium bromide was obtained by adding 1.2:1 molar ratio of n-butyl bromide and n-butyl imidazole at 90 °C for 12 h, unreacted reactants are removed under reduced pressure [29]. The obtained ionic liquid [DBIm]Br structure was confirmed by ¹HNMR spectroscopy. The conductivity of 1,3-dibutylimidazolium bromide [DBIm]Br was observed up to 0.127 mS/cm at 25 °C.

Scheme 1

Later, the ionic liquid electrolyte was prepared by mixing anhydrous Mg(ClO₄)₂ salt and 1,3 dibutyl imidazolium bromide [DBIm]Br in an argon filled with the glove box (MBRAUN, Inc.). The molar ratio of Mg(ClO₄)₂ and [DBIm]Br was kept at 1:4, in which the ionic conductivity was increased to 0.275 mS/cm at 25 °C.

2.5 construction of two-electrode cells

Two-electrode cells were constructed in a glove box (MBRAUN, Germany) filled with an argon gas (99.999%). The working electrodes were prepared by slurry (such as 5% PVDF in NMP solvent, 90% active material and 5% super-p carbon black) coating method. Mg electrode used as a reference and counter electrode and a Celgard film was used as a separator. The electrolyte was $\text{Mg}(\text{ClO}_4)_2$ in an ionic liquid [DBIm]Br and constructed in a closed container

2.6 Characterization techniques

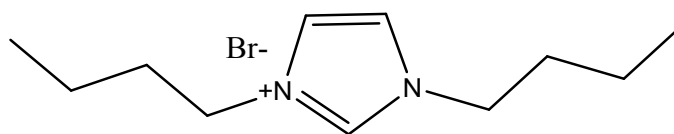
The physical properties of the exfoliated and fluorinated products were investigated by X-ray photoelectron spectroscopy (XPS, Kratos-Axis Supra), field emission scanning electronic microscope (FE-SEM, Hitachi– S520)), energy dispersive x-ray spectroscopy (EDX) and X-ray diffraction (XRD, Hitachi with field emission gun). The ^1H NMR spectrum of the ionic liquid [DBIm]Br in DMSO- d_6 is operated on Avance 300 proton NMR spectrometer. Electrochemical measurements such as cyclic voltammetry, electrochemical impedance spectroscopy and charge-discharge studies of a cell were performed on Bio-logic SAS model 1-VSP and NEWARE BST-5V electrochemical workstation.

3. Results and discussions

3.1 Proton NMR spectroscopy (PNMR)

The structure of ionic liquid [DBIm]Br is conformed by proton NMR spectrum (Fig. 2). The chemical shifts between 0.9, 1.4 and 1.9 ppm conforming to the two methyl (CH_3) and two ethyl (CH_2) protons of alkyl chains. The chemical shift at around 4.38 ppm corresponds to two N-methyl (N-CH_2) protons. The imidazole ring protons resonate in an aromatic region to the around 7.0 to 10.5 ppm, respectively.

The structure of the ionic liquid [DBIm]Br is given below

**Figure 2**

3.2. X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS) analysis was conducted to confirm the presence of F 1s in fluorinated h-BNNSs. From Fig. 3, it is observed that the binding energy (BE) values of B 1s and N 1s of the h-BN compound are in good acceptance with the report values [31]. The bulk h-BN is a major peak in B 1s XPS located at 190.38 eV [31] represents a high binding energy position of B atoms. The F-h-BNNSs peak signal in B 1s XPS located at lower binding energy position around 188.68 eV represents F atom binding with sp^2 B atoms. The bulk h-BN peak signal in N 1s XPS located at 398.10 eV [30] represents high binding energy position of N atoms. The F-h-BNNSs peak signal in N 1s XPS located at the lower binding energy around 396.01 eV represents F atom unbounded with N atoms, due to electron donating N atom which is can be associated in the formation of few layers of h-BN. The F-h-BNNSs peak signal in F 1s XPS located at binding energy near 684.63 eV can be associated with the F atom in B-N-B bonds. The binding energy 684.63 eV value for F 1s in boron fluoride is similar to ionic F in LiF (F1s:684.1 eV) rather than covalent character F in Teflon (F 1s: 689.1 eV) [31].

Figure 3

3.3. FE-SEM analysis

The morphologies of bulk h-BN (B-h-BN) powder and exfoliated h-BN were examined using FE-SEM. Fig. 4 (a-b) demonstrate the low to high-magnification FE-SEM model of the bulk h-BN powder. The low-magnification SEM model in Fig. 4(a) represents the whole morphology of the bulk BN powder. In Fig. 4(b) shows that the bulk BN powder consists of

large-scale flakes and some of the thick flakes with sizes of the range about 10 μm . Low to high-magnification FE-SEM image of prepared exfoliated h-BN using the isopropyl alcohol (IPA) as a solvent. Fig. 4(c-d) reveals that blooming flower-like morphology and many sheets are stacked together. It is clearly observed that the sheets have a diameter of 1 μm and sheets edge thickness of about 100 nm. The exfoliated h-BNNSs attained laminated structure and decreasing size compared to bulk h-BN. Therefore, it can be concluded that the thickness of the bulk h-BN has been reduced, implying the peeled-off structure of h-BNNSs from the bulk h-BN.

Figure 4

3.4. EDX spectrum

EDX analysis is recorded in Fig. 5, which clearly shows the strong signals of B and N, and a weak signal of F from the fluorinated h-BNNSs sample. The respective atomic wt. % of N, B, and F are observed as 53.5 %, 45.9 % and 0.6 % for F-h-BNNSs sample respectively as shown in Fig. 5.

Figure 5

3.5. X-ray studies of the fluorinated h-BNNSs, exfoliated h-BN and h-BN

Fig. 6 shows comparative XRD patterns of h-BN, h-BNNSs, and F-h-BNNSs. It illustrates the diffraction peak of F-h-BNNSs, h-BNNSs, and h-BN corresponded to (002) plane of the hexagonal phase of BN, with lattice constants of $a = b = 2.504$ and $c = 6.656$ Å. (JCPDS card 34-0421). X-ray diffraction is very helpful for the information about interlayer spacing in the corresponding diffraction angle. The (002) crystal plane of h-BN crystal similar to the (002) plane of graphite. However, it is observed that the h-BN gave a corresponding peak at 26.84° and h-BNNSs at 26.78° (due to (002) plane) which affected its crystal lattice and the interlayer

spacing since it increased from 3.32 to 3.325 Å, which represents the (002) crystal faces of h-BNNSs are vulnerable more than h-BN powder, and h-BNNSs may be exfoliated from the bulk BN powder along with (002) plane [32]. The fluorination process explains the defect sites in the crystal lattice. Fig. 6 shows the h-BNNSs gave a corresponding peak at 26.78° and F-h-BNNSs at 26.75° (due to (002) plane) which defect its crystal lattice and the interlayer spacing since it increased from 3.325 to 3.328 Å, indicating that F atoms were doped into the layered BNNSs structure [33, 34].

Figure 6

3.6. Cyclic voltammetry studies with two electrode cell assembly

The electrochemical behaviour of h-BNNSs and F-h-BNNSs were studied using cyclic voltammetry with 0.5M $\text{Mg}(\text{ClO}_4)_2$ in the imidazole-based ionic liquid electrolyte as shown in Fig. 7. CV is very useful for information about intercalation/de-intercalation of corresponding cathode material in magnesium batteries. The electrochemical behaviour of two materials is observed to be in the potential window of 0 to 3V. However, it is absorbing that the F-h-BNNSs redox reaction in the potential range and shows the stable reversibility behaviour within the 5 cycles. The F-h-BNNSs shows the anodic peak at 1.75 V can be considered as the intercalation of the Mg^{2+} ion on the edge sites of BN sheets and corresponding cathodic peak at 0.8 V can be associated with the de-intercalation of the Mg^{2+} ion on the surface of BN layers [35], the current density calculated from CV was 48 mA g^{-1} . However, it is absorbing that the h-BNNSs shows no redox reaction occurs in the potential range that means difficulty Mg-ions intercalation and de-intercalation on the surface of BN sheets because of semiconductivity property of BN. The current density was calculated from CV and found 20 mA g^{-1} . The CV results considered that the reversibility and the current density of F-h-BNNSs are better than the h-BNNSs.

Figure 7*3.7. Electrochemical impedance studies (EIS)*

The surface resistivity of the two cells was studied using EIS. Fig. 8 refers to the Nyquist plots in the frequency range 10 MHz to 1000 MHz. All Nyquist plots were recorded from high frequency to low-frequency region, the semi-circle in a high-frequency region is assigned to the solid electrolyte interface (SEI) film between electrode and electrolyte, and the semi-circle in medium frequency corresponding to charge transfer resistance, and a straight line at low-frequency region corresponds to the Warburg impedance corresponding to ion diffusion resistance, respectively. The corresponding EIS circuit model of handling system is shown in the inset of Fig. 8. The symbol R_1 can be assigned to an internal resistance of two samples, R_{SEI} and R_{ct} can be represented as SEI resistance and the charge transfer resistance respectively, C_{dl} and Z_w can be corresponding to the double layer capacitances and the Warburg impedance. Fig. 8 shows that the h-BNNSs has maintained R_{SEI} resistance of 128 Ω , which represents poor electrochemical performance. Further, R_{SEI} resistance of F-h-BNNSs was decreased to 94 Ω which gave a good electrochemical performance. The R_{ct} resistance of h-BNNSs and F-h-BNNSs are 209 Ω and 137 Ω , respectively. It is appropriate to conclude that the F-h-BNNSs conductivity tremendously increased.

Figure 8*3.8. Charge-discharge studies of the cells*

The electrochemical behaviour of h-BNNSs and F-h-BNNSs for magnesium battery system was further evaluated by galvanostatic charge/discharge studies at the current density of 15 mA/g. The charge-discharge profiles of two cells are shown in the Fig. 9(a). The F-h-BNNSs discharge rate started at high potential maintaining a good plateau at 1.35 V, which

represents the magnesium ion intercalation on the surface edges of BN sheets [35] and the h-BNNSs discharge rate started at low voltage plateau at 1.1 V, this represents the difficulty Mg-ion intercalation on the surface of BN sheets. The F-h-BNNSs has a good discharge capacity of 50 mAhg⁻¹ than h-BNNSs (40 mAhg⁻¹). The coulombic efficiency of the h-BNNSs and F-h-BNNSs is about 60 %, and 65 % respectively during the initial charge/discharge cycle and the coulombic efficiency improved by increasing the charge/discharge cycles. The galvanotactic charge/discharge cycling (80 cycles) performances of fluorinated h-BNNSs/Mg cell. As shown in Fig. 9(b) also shows the cycling performance of F-h-BNNSs and h-BNNSs, the discharge capacity of F-h-BNNSs shows much good capacity around 50 mAh/g within 80 cycles and the result indicates that the capacity increased with increased cycles. The discharge capacity of h-BNNSs shows much poor capacity around 40 mAh/g within 22 cycles due to semiconductivity property of BN. Fig. 9(c) shows the voltage vs time plots of the F-h-BNNSs. It is observed that the discharge potential plateau is improved by increasing the charge/discharge cycles because of its Mg²⁺ ion interest to insertion of the surface of the BN layers.

Figure 9

4. Conclusion

In summary, we have established the fluorine (F) doping into the nanosheets of boron nitride through exfoliation BN from bulk BN powder in isopropyl alcohol solution via chemical solution process with fluoroboric acid. A good electrical conductivity of the exfoliated and fluorinated h-BNNSs is obtained. The electrochemical performance was improved by using fluorination method. The fluorinated h-BNNSs as cathode material for rechargeable Mg battery. Which exhibits a good discharge capacity of 50 mAhg⁻¹, with good rate capability and extremely good cycling stability.

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Figure 1

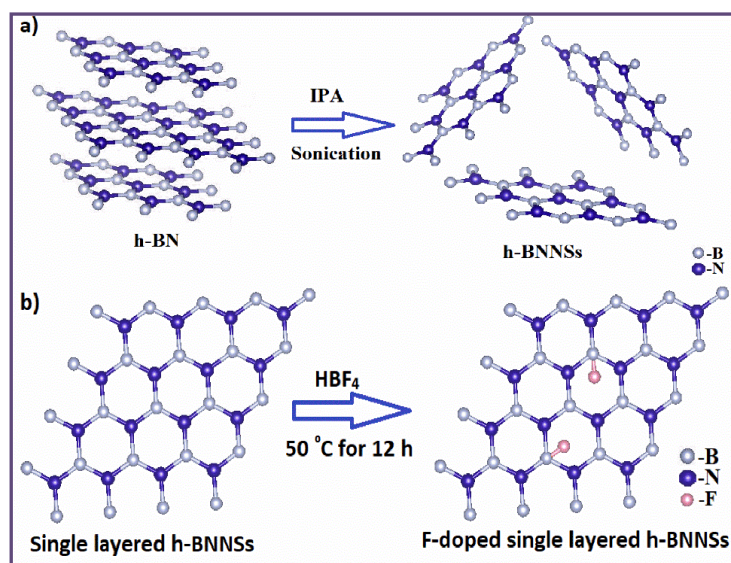
**Fig.1** Schematic diagram for the preparation of (a) h-BNNSs and (b) F-h-BNNSs.

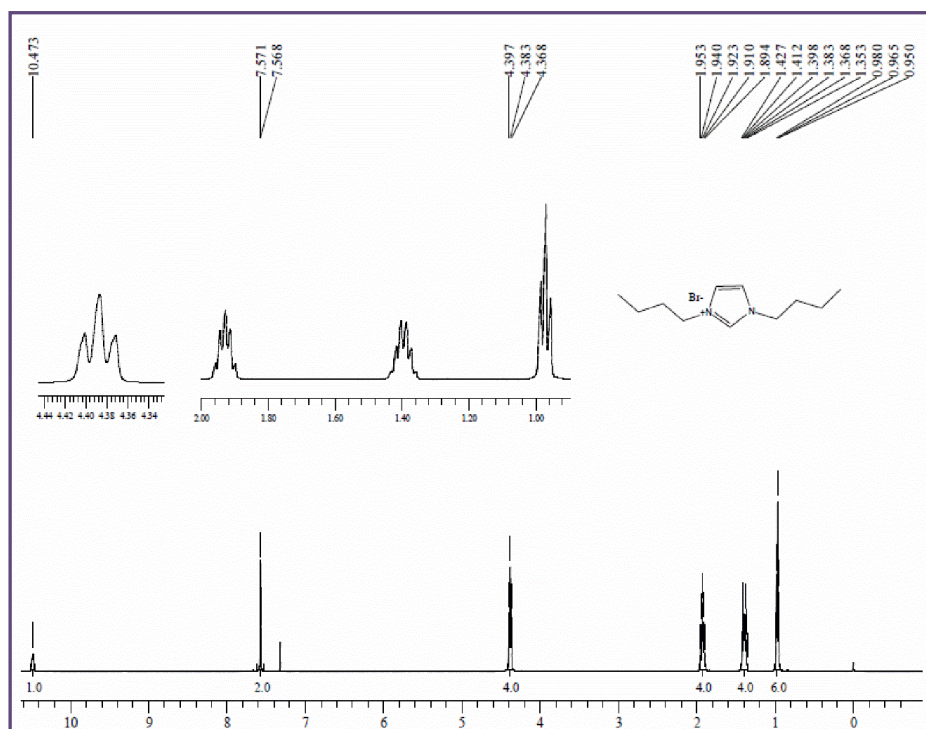
Figure 2**Fig.2** PNMR spectrum of N,N-di-n-butylimidazolium bromide

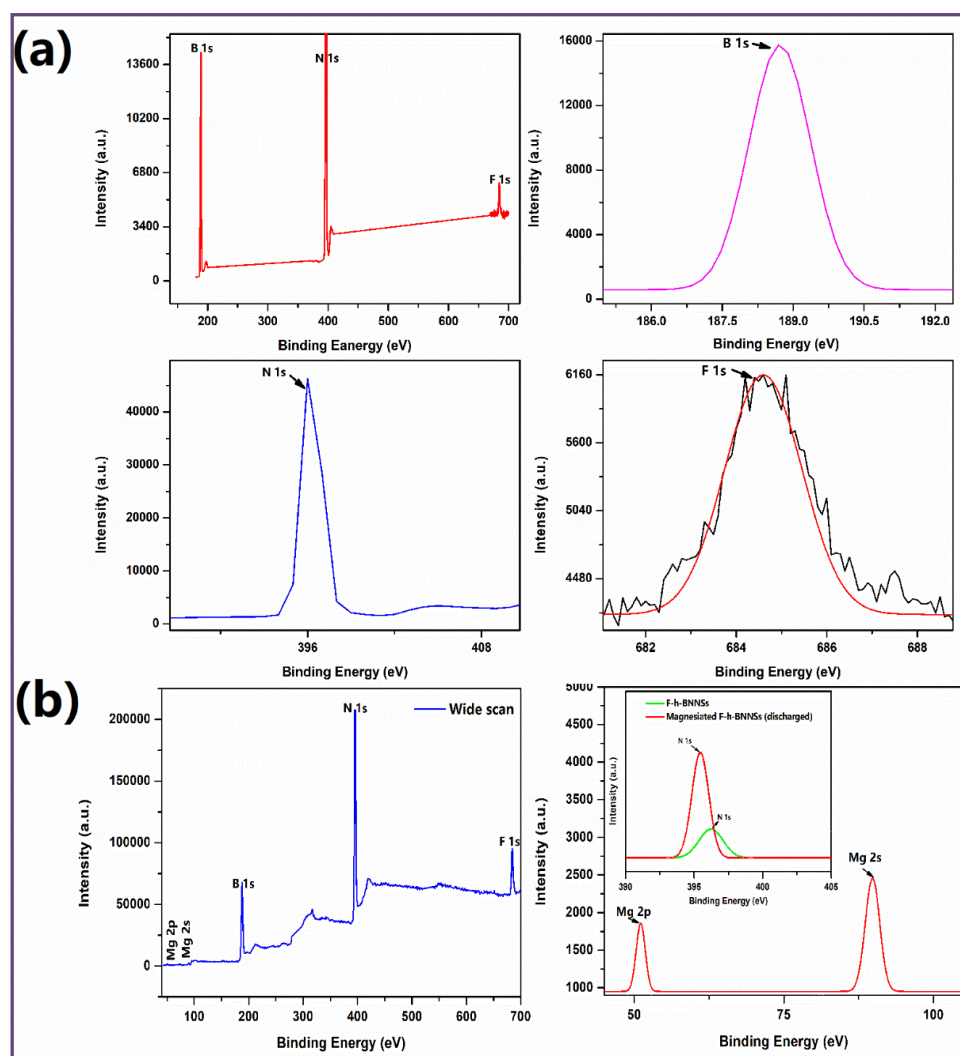
Figure 3**Fig. 3** X-ray photoelectron spectroscopy recorded for F-h-BNNSs (a) and magnesiated F-h-BNNSs (discharged) (b).

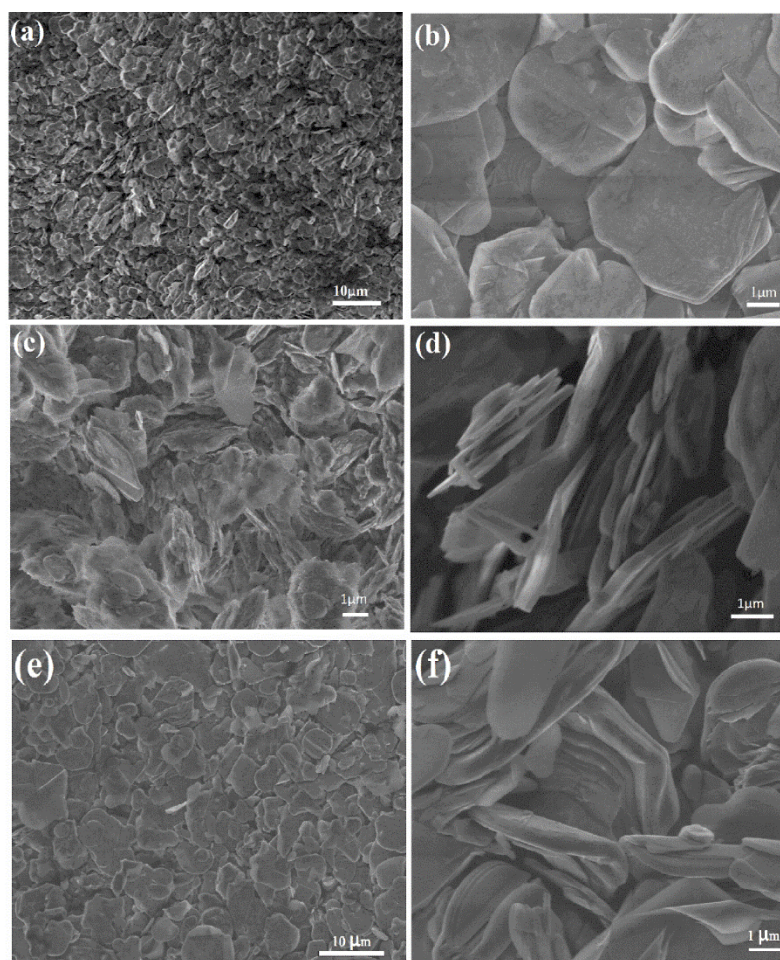
Figure 4**Fig. 4** FE-SEM image of bulk h-BN (a-b), exfoliated h-BN (c-d) and F-h-BNNSs (e-f)

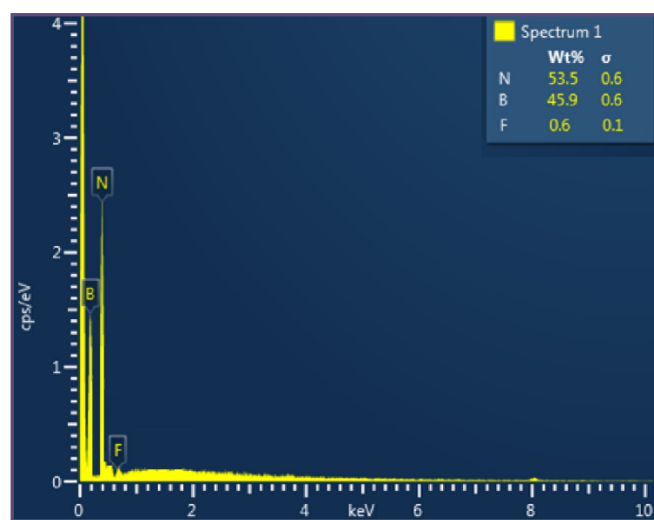
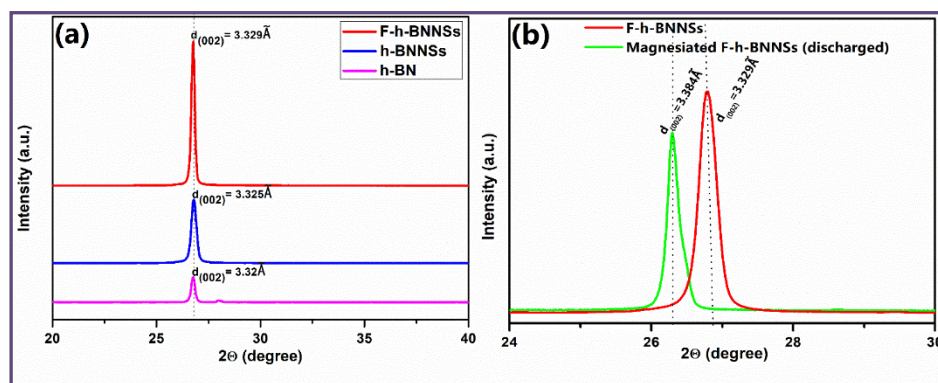
Figure 5**Fig. 5** EDX spectrum of fluorinated h-BNNSs (F-h-BNNSs).

Figure 6**Fig. 6** (a) XRD pattern of bulk h-BN, h-BNNSs, and F-h-BNNSs.

(b) XRD pattern of F-h-BNNSs and magnesianated F-h-BNNSs (discharged)

Figure 7

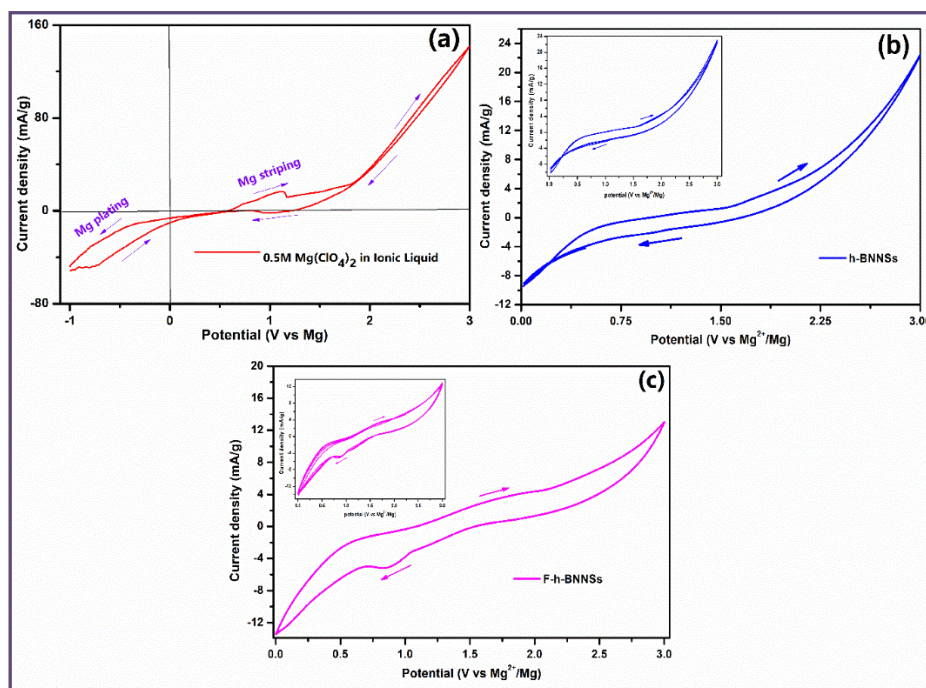


Fig. 7 (a) CV of the reversible Mg stripping/plating using 0.5M $\text{Mg}(\text{ClO}_4)_2$ in ionic liquid electrolyte (Pt strip as WE and Mg as CE & RE) at 5mVs^{-1} scan rate.

CV plots at scan rate of 5mVs^{-1} in the voltage range of 0.0 – 3.0 V; (b) h-BNNSs and (c) F-h-BNNSs.

Figure 8

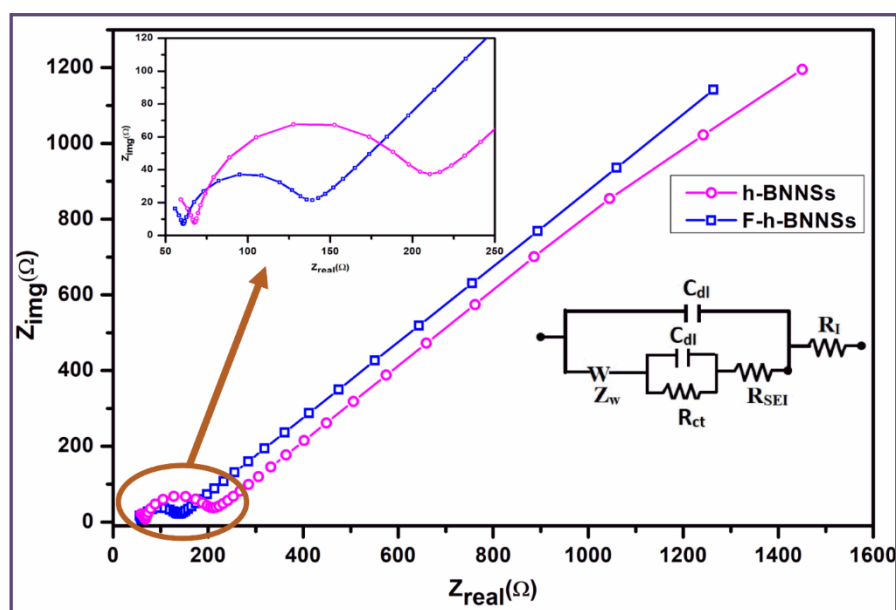


Fig. 8 Impedance spectrum of h-BNNSs and F-h-BNNSs.

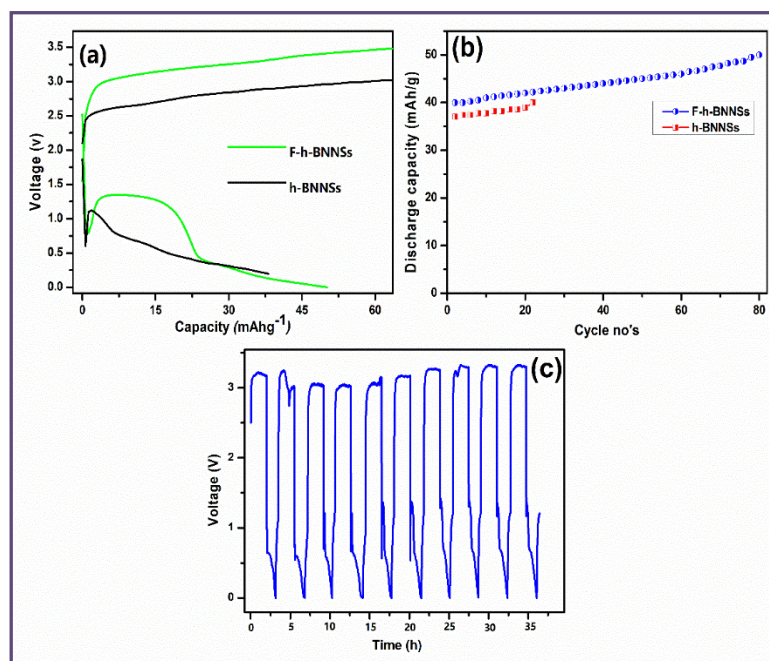
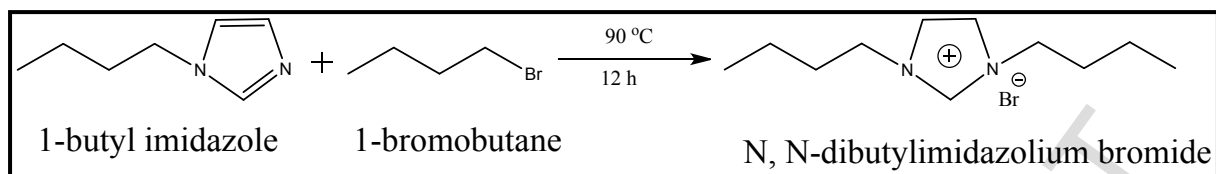
Figure 9

Fig. 9 (a) Voltage vs. capacity of h-BNNSs and F-h-BNNSs (b) cycle life of h-BNNSs and F-h-BNNSs (c) voltage vs time of F-h-BNNSs.

Scheme 1

Scheme 1 Represents the synthesis of N, N-dibutylimidazolium bromide.

Highlights

- F doped h-BNNSs were prepared from a chemical process using HBF_4
- F doped h-BNNSs edge thickness was found at around 100 to 150 nm.
- The obtained F doped h-BNNSs exhibited good capacity and rate capability.
- F doped h-BNNSs shown a discharge capacity of 50 mAhg^{-1} within 80 cycles.

Table 1

Table 1. Impedance fitted parameters are derived from equivalent EIS circuit model for h-BNNSs and F-h-BNNSs electrodes

Electrode	$R_I (\Omega)$	$R_{SEI} (\Omega)$	$C_{dl} (F)$	$R_{ct} (\Omega)$	$C_{dl} (F)$
h-BNNSs	66	128	0.840×10^{-9}	209	0.362×10^{-6}
F-h-BNNSs	58	94	0.72×10^{-9}	137	0.292×10^{-6}